

Lecture Notes: Functions

Discrete Mathematics

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Section 1 Functions

Definition (Function). Let A and B be non-empty sets. A *function* (or *mapping*) f from A to B , denoted $f : A \rightarrow B$, is a rule that assigns to each element $x \in A$ exactly one element $y \in B$. The element y is called the *image* of x under f and is written $f(x)$.

Remark. A function is a special type of binary relation: it is a relation $f \subseteq A \times B$ such that for every $a \in A$, there exists exactly one $b \in B$ with $(a, b) \in f$. This is sometimes called a "well-defined" assignment.

1.1 Domain, Codomain, and Range

- ▷ **Domain:** The set A is called the *domain* of f , denoted $\text{dom}(f)$. It is the set of all possible inputs.
- ▷ **Codomain:** The set B is called the *codomain* of f . It is the set within which all outputs lie.
- ▷ **Range (Image):** The *range* or *image* of f , denoted $\text{ran}(f)$ or $f(A)$, is the set of all actual outputs:

$$\text{ran}(f) = \{f(x) \mid x \in A\} \subseteq B.$$

Remark. Note carefully: the codomain is part of the definition of the function, while the range is a derived set that is always a subset of the codomain.

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = x^2$.

- Domain: $\mathbb{R}$.
- Codomain: $\mathbb{R}$.
- Range: $[0, \infty) = \{y \in \mathbb{R} \mid y \geq 0\}$.

Notice that the range is a proper subset of the codomain.

Consider $g : \{1, 2, 3\} \rightarrow \{a, b, c, d\}$ defined by

$$g(1) = a, \quad g(2) = c, \quad g(3) = a.$$

Then $\text{dom}(g) = \{1, 2, 3\}$, codomain is $\{a, b, c, d\}$, and $\text{ran}(g) = \{a, c\}$.

1.2 Function Notation and Equality

We write $f : A \rightarrow B$ to denote a function, and for $x \in A$, the output is denoted $f(x)$.

Definition (Equality of Functions). Two functions $f : A \rightarrow B$ and $g : C \rightarrow D$ are *equal*, written $f = g$, if and only if:

- (i) $A = C$ (same domain),
- (ii) $B = D$ (same codomain),
- (iii) $f(x) = g(x)$ for all $x \in A$ (same rule).

Definition (Image of a Set). For a subset $X \subseteq A$, the *image* of X under f is

$$f(X) = \{f(x) \mid x \in X\}.$$

For a subset $Y \subseteq B$, the *preimage* (or *inverse image*) of Y is

$$f^{-1}(Y) = \{x \in A \mid f(x) \in Y\}.$$

Section 2 Types of Functions

Functions can be classified based on how they map elements from the domain to the codomain. The three most important classes are injective, surjective, and bijective functions.

2.1 One-to-One (Injective) Functions

Definition (Injective Function). A function $f : A \rightarrow B$ is *injective* (or *one-to-one*) if distinct elements of the domain map to distinct elements of the codomain. Formally:

$$\forall x_1, x_2 \in A, (f(x_1) = f(x_2) \implies x_1 = x_2).$$

Equivalently (contrapositive): $x_1 \neq x_2 \implies f(x_1) \neq f(x_2)$.

Intuition: In an injective function, no two different inputs produce the same output. Each element of the range is hit by at most one element from the domain.

[Injective] The function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 3x + 2$ is injective. Indeed, if $3x_1 + 2 = 3x_2 + 2$, then $3x_1 = 3x_2$, so $x_1 = x_2$.

[Not Injective] The function $g : \mathbb{R} \rightarrow \mathbb{R}$ given by $g(x) = x^2$ is **not** injective, because $g(2) = 4$ and $g(-2) = 4$, yet $2 \neq -2$.

2.2 Onto (Surjective) Functions

Definition (Surjective Function). A function $f : A \rightarrow B$ is *surjective* (or *onto*) if every element of the codomain B is the image of at least one element of the domain A . Formally:

$$\forall y \in B, \exists x \in A \text{ such that } f(x) = y.$$

Equivalently, the range equals the codomain: $\text{ran}(f) = B$.

Intuition: A surjective function "covers" the entire codomain: every element in B is "hit" by the function.

[Surjective] The function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = x^3$ is surjective. For any $y \in \mathbb{R}$, we can take $x = \sqrt[3]{y}$ (a real number) and $f(x) = y$.

[Not Surjective] The function $g : \mathbb{R} \rightarrow \mathbb{R}$ given by $g(x) = x^2$ is **not** surjective, because there is no real x such that $g(x) = -1$ (the range is only $[0, \infty)$).

2.3 Bijective Functions

Definition (Bijective Function). A function $f : A \rightarrow B$ is *bijective* (or a *one-to-one correspondence*) if it is both injective and surjective. That is, every element of B is the image of exactly one element of A .

Intuition: A bijection pairs up elements of A and B uniquely in both directions. This is the key concept for comparing the "sizes" of sets.

[Bijective] The function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = 2x + 1$ is bijective.

– Injective: $2x_1 + 1 = 2x_2 + 1 \implies x_1 = x_2$.

– Surjective: For any $y \in \mathbb{R}$, $x = \frac{y-1}{2}$ satisfies $f(x) = y$.

[Bijective on Restricted Domain] The function $h : [0, \infty) \rightarrow [0, \infty)$ defined by $h(x) = x^2$ is bijective. It is injective because on non-negative numbers, $x_1^2 = x_2^2$ implies $x_1 = x_2$; it is surjective because every non-negative y has a non-negative square root $x = \sqrt{y}$.

Section 3 Inverse Functions

Definition (Inverse Function). Let $f : A \rightarrow B$ be a bijective function. The *inverse function* of f , denoted $f^{-1} : B \rightarrow A$, is the function that assigns to each $y \in B$ the unique $x \in A$ such that $f(x) = y$. That is,

$$f^{-1}(y) = x \quad \text{if and only if} \quad f(x) = y.$$

Theorem (Existence of Inverse). A function $f : A \rightarrow B$ has an inverse function $f^{-1} : B \rightarrow A$ if and only if f is bijective.

Proof Sketch. If f is not injective, then $f(x_1) = f(x_2) = y$ for some $x_1 \neq x_2$, making it impossible to define $f^{-1}(y)$ uniquely. If f is not surjective, then some $y \in B$ has no preimage, so $f^{-1}(y)$ is undefined. Conversely, if f is bijective, each y has exactly one x with $f(x) = y$, defining a proper function.

3.1 Finding Inverse Functions

To find the inverse of a bijective function f , solve the equation $y = f(x)$ for x in terms of y , then interchange the roles of x and y (or simply express $f^{-1}(x)$).

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = 3x - 7$. Find f^{-1} .

Solution. Set $y = 3x - 7$. Solve for x :

$$\begin{aligned} y &= 3x - 7 \\ y + 7 &= 3x \\ x &= \frac{y + 7}{3}. \end{aligned}$$

Thus, $f^{-1}(y) = \frac{y+7}{3}$, or equivalently, $f^{-1}(x) = \frac{x+7}{3}$.

Let $g : [0, \infty) \rightarrow [0, \infty)$ be $g(x) = x^2 + 1$. Find g^{-1} .

Solution. Set $y = x^2 + 1$. Solve for x with $x \geq 0$:

$$\begin{aligned} y - 1 &= x^2 \\ x &= \sqrt{y - 1} \quad (\text{taking the non-negative root since } x \geq 0). \end{aligned}$$

Thus, $g^{-1}(x) = \sqrt{x - 1}$, with domain $[1, \infty)$ (which matches $\text{ran}(g)$).

3.2 Verification by Composition

Theorem (Cancellation Laws). If $f : A \rightarrow B$ is bijective and $f^{-1} : B \rightarrow A$ is its inverse, then:

$$\begin{aligned} f^{-1}(f(x)) &= x \quad \text{for all } x \in A, \\ f(f^{-1}(y)) &= y \quad \text{for all } y \in B. \end{aligned}$$

This provides a practical way to verify an inverse: compose the functions in both orders and check that the identity results.

For $f(x) = 3x - 7$ and $f^{-1}(x) = \frac{x+7}{3}$:

$$\begin{aligned} f^{-1}(f(x)) &= f^{-1}(3x - 7) = \frac{(3x - 7) + 7}{3} = \frac{3x}{3} = x, \\ f(f^{-1}(x)) &= f\left(\frac{x + 7}{3}\right) = 3\left(\frac{x + 7}{3}\right) - 7 = x + 7 - 7 = x. \end{aligned}$$

Section 4 Composition of Functions

Definition (Composition). Let $f : A \rightarrow B$ and $g : B \rightarrow C$ be functions. The *composition* of g with f , denoted $g \circ f$ (read "g after f" or "g of f"), is the function $g \circ f : A \rightarrow C$ defined by

$$(g \circ f)(x) = g(f(x)) \quad \text{for all } x \in A.$$

Remark (Domain Considerations). For $g \circ f$ to be defined, the codomain of f must equal (or be a subset of) the domain of g . The diagram is:

$$\begin{array}{ccccc} A & \xrightarrow{f} & B & \xrightarrow{g} & C \\ & & A & \xrightarrow{g \circ f} & C \end{array}$$

4.1 Properties of Composition

Theorem (Associativity). Composition of functions is associative. If $f : A \rightarrow B$, $g : B \rightarrow C$, and $h : C \rightarrow D$, then

$$(h \circ g) \circ f = h \circ (g \circ f).$$

Proof. For any $x \in A$:

$$\begin{aligned} ((h \circ g) \circ f)(x) &= (h \circ g)(f(x)) = h(g(f(x))) \\ (h \circ (g \circ f))(x) &= h((g \circ f)(x)) = h(g(f(x))). \end{aligned}$$

Both are equal, so associativity holds.

Remark (Non-Commutativity). Composition of functions is **not** commutative in general. Even when both orders are defined, $f \circ g$ and $g \circ f$ are usually different.

Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be given by $f(x) = x^2$ and $g(x) = x + 1$. Then:

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) = f(x + 1) = (x + 1)^2 = x^2 + 2x + 1, \\ (g \circ f)(x) &= g(f(x)) = g(x^2) = x^2 + 1. \end{aligned}$$

Clearly, $(f \circ g)(x) \neq (g \circ f)(x)$ (for example, evaluate at $x = 0$).

4.2 Composition with Inverse

Theorem. If $f : A \rightarrow B$ is bijective, then

$$\begin{aligned} f^{-1} \circ f &= \text{id}_A \quad (\text{the identity function on } A), \\ f \circ f^{-1} &= \text{id}_B \quad (\text{the identity function on } B). \end{aligned}$$

This property is often used to prove that a function is bijective by constructing its inverse.

4.3 Composition and Function Types

Theorem. Let $f : A \rightarrow B$ and $g : B \rightarrow C$.

- (i) If f and g are both injective, then $g \circ f$ is injective.
- (ii) If f and g are both surjective, then $g \circ f$ is surjective.
- (iii) If f and g are both bijective, then $g \circ f$ is bijective.

Section 5 Cardinality of Sets

Definition (Cardinality). The *cardinality* of a set A , denoted $|A|$, is a measure of the "number of elements" in A . For finite sets, it is simply the count of distinct elements.

Definition (Finite and Infinite Sets).

- A set is *finite* if it has n elements for some $n \in \mathbb{N} \cup \{0\}$; then $|A| = n$.
- A set is *infinite* if it is not finite.

5.1 Comparing Cardinalities via Functions

Functions, especially bijections, provide a rigorous way to compare the sizes of sets without counting elements explicitly.

Definition (Equinumerosity). Two sets A and B have the same *cardinality* (are *equinumerous* or *equipotent*), denoted $|A| = |B|$, if there exists a bijection $f : A \rightarrow B$.

Theorem (Cantor—Bernstein—Schröder). If there exists an injection $f : A \rightarrow B$ and an injection $g : B \rightarrow A$, then there exists a bijection $h : A \rightarrow B$, so $|A| = |B|$. (We state this without proof here.)

Let $A = \{1, 2, 3, 4\}$ and $B = \{a, b, c, d\}$. The function $f : A \rightarrow B$ defined by $f(1) = a, f(2) = b, f(3) = c, f(4) = d$ is a bijection, so $|A| = |B| = 4$.

[Even and Natural Numbers] Let $\mathbb{N} = \{1, 2, 3, \dots\}$ and $E = \{2, 4, 6, \dots\}$. The function $f : \mathbb{N} \rightarrow E$ defined by $f(n) = 2n$ is a bijection. Therefore, $|\mathbb{N}| = |E|$, even though $E \subset \mathbb{N}$. This demonstrates that an infinite set can have the same cardinality as a proper subset of itself — a defining characteristic of infinite sets (Dedekind's definition).

5.2 Countable and Uncountable Sets

Definition (Countable Set). A set A is *countable* if it is finite or if there exists a bijection $f : \mathbb{N} \rightarrow A$ (equivalently, $|A| = |\mathbb{N}|$, denoted $\aleph_0$, "aleph-null"). In this case, the elements can be listed as $a_1, a_2, a_3, \dots$.

Examples of Countable Sets:

- $\mathbb{N}, \mathbb{Z}, \mathbb{Q}$ (rational numbers) are countably infinite.
- Any finite set is countable by definition.

Theorem (Cantor's Diagonal Argument). The set of real numbers $\mathbb{R}$ is *uncountable*. That is, there is no bijection from $\mathbb{N}$ to $\mathbb{R}$. The cardinality of $\mathbb{R}$ is strictly greater than $\aleph_0$, often denoted $\mathfrak{c}$ (the continuum).

[Countability of $\mathbb{Z}$] We can show $|\mathbb{Z}| = |\mathbb{N}|$ by constructing an explicit bijection $f : \mathbb{N} \rightarrow \mathbb{Z}$:

$$f(n) = \begin{cases} \frac{n}{2} & \text{if } n \text{ is even,} \\ -\frac{n-1}{2} & \text{if } n \text{ is odd.} \end{cases}$$

The listing order is: $0, 1, -1, 2, -2, 3, -3, \dots$ (if we consider $\mathbb{Z}$ including 0, with $\mathbb{N}$ shifted appropriately). This demonstrates that $\mathbb{Z}$ is countably infinite.

[Uncountability of the Interval $(0, 1)$] The open interval $(0, 1)$ is uncountable. Any attempted list $r_1, r_2, r_3, \dots$ of real numbers in $(0, 1)$, expressed in decimal expansion, can be shown to miss some number by constructing a new real number that differs from r_k in the k -th decimal place (Cantor's diagonal argument). Hence, $(0, 1)$ is not countable.

Remark. The concept of cardinality is a profound extension of counting to infinite sets. The study of infinite cardinalities leads to the hierarchy:

$$|\mathbb{N}| = \aleph_0 < |\mathbb{R}| = \mathfrak{c} < |\mathcal{P}(\mathbb{R})| < \dots$$

A fundamental question (the Continuum Hypothesis) asks whether there exists a set with cardinality strictly between $\aleph_0$ and $\mathfrak{c}$.